

Cardiovascular Disease Around the World

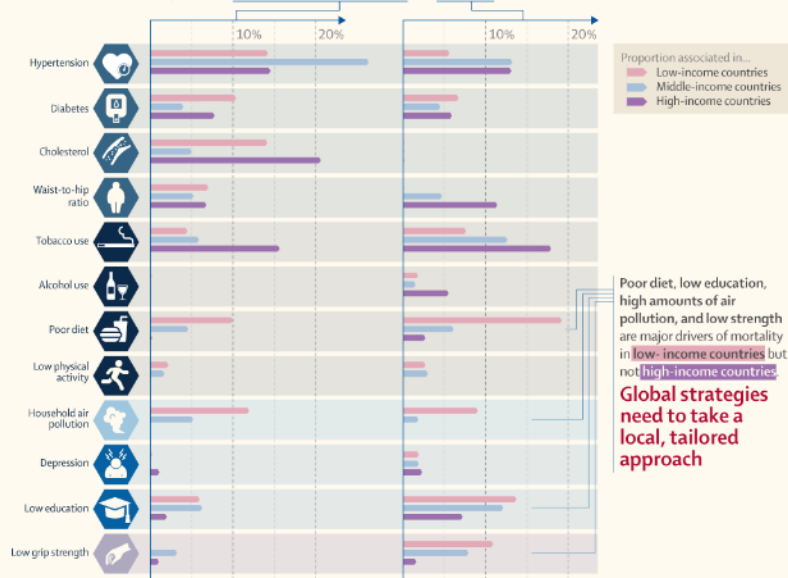


Over 70% of cardiovascular disease worldwide is caused by modifiable risk factors



But the causes are not the same around the world

Proportion of cardiovascular disease cases and mortality attributable to risk factors



GLOBAL HEARTS INITIATIVE
Working together to beat cardiovascular disease

THE GLOBAL CARDIOVASCULAR DISEASE (CVD) CRISIS

Cardiovascular disease is the world's biggest killer, claiming **17.5 million** lives per year, 1/3 prematurely under 70 years.

This number is steadily increasing

80% of all CVD deaths are from heart attacks and strokes

75% occur in developing countries

Premature deaths from heart attacks and strokes threaten global health and development.

RISK FACTORS

Major risk factors contributing to CVDs are:



TOBACCO USE



CONSUMPTION OF FOODS HIGH IN SALT



HIGH BLOOD PRESSURE

THE GLOBAL HEARTS INITIATIVE: Responding to the global cardiovascular disease crisis

An initiative to scale up national responses for prevention and management of cardiovascular diseases.



Technical package to defeat the global tobacco epidemic



Technical package for salt reduction



Technical package for cardiovascular disease management in primary health care

In the 2030 Sustainable Development Agenda, governments have committed to step up action to reduce premature deaths from cardiovascular disease and other NCDs to improve health and promote development.

www.who.int/global_hearts



For more, visit www.thelancet.com

- Modifiable risk factors, cardiovascular disease, and mortality in 155 722 individuals from 21 high-income, middle-income, and low-income countries (PURE)
- Variations in common diseases, hospital admissions, and deaths in middle-aged adults in 21 countries from five continents (PURE): a prospective cohort study

THE LANCET

The best science for better lives

Leading Causes of Cardiovascular Disease (CVD) Deaths

1. Ischemic Heart Disease
2. Stroke
3. Diabetes Mellitus
4. Hypertensive Heart Disease

Behavioral Risk Factors of CVDs

- Unhealthy diet
- Physical inactivity
- Tobacco use
- Harmful use of alcohol



Big Picture: 2026 AHA Dietary Guidance

- Focus on the overall dietary pattern across the life course
- Eat a wide variety of vegetables and fruits
- Choose mostly whole grains and healthy protein sources
- Replace saturated fats with unsaturated fats
- Choose minimally processed foods over ultra-processed foods
- Minimize added sugars and sodium
- If alcohol is not consumed, do not start; if consumed, limit intake

CVD		Prevention	Treatment
		INCREASE Vegetables Legumes Fish Fruits (<i>limited</i>) Olive oil (80 gram/day) Whole grain (< 100 gram/day) Nuts (30 gram/day) Time-restricted eating (11- 12 hours) Periodic Fasting-mimicking diet Exercise AVOID Red meat (<i>particularly processed</i>) Sugar (> 10 gram/day) Dairy Protein (> 0.36 gram/pound of bodyweight until age 65-70) Saturated and Trans fats Meal frequency (> 3 meals/day) Overweight/Obesity Stress	Longevity diet: ➤ Vegan/pescetarian-based diet ➤ No dairy ➤ Low fruit and fruit-juices ➤ Yes fish and extra-virgin olive oil ➤ Yes grains, legumes, lentils, vegetables ➤ Yes nuts (walnuts, hazelnuts, almonds) ➤ No caloric restriction (except overweight/ obese) ➤ Time-restricted eating (11-12 hours). ➤ 60% carbohydrate, 30% fat, 10% protein Ornish diet: ➤ Vegetarian diet ➤ 10% of calories as fat, 15-20% protein, 70-75% predominantly complex carbohydrates ➤ No caloric restriction ➤ Lasting > 1 year Esselstyn diet: ➤ Vegetarian diet ➤ No fish, oil, or dairy ➤ Consume grains, legumes, lentils, vegetables, and fruit ➤ No caloric restriction



Saturated vs. Trans-Fat Effect on Lipid Panel

- Increases **total** and **LDL-C** cholesterol and decreases **HDL**
- Effect of Dietary trans Fatty Acids on High-Density and Low-Density Lipoprotein Cholesterol Levels in Healthy Subjects*

	Trans fat	Saturated fat
Total cholesterol	+6%	+12%
LDL cholesterol	+14%	+18%
HDL cholesterol	-12%	0%
LDL/HDL ratio	+29%	+18%

Omega-3 Polyunsaturated Fatty Acids (PUFAs)

EPA and DHA

- Prefer fish and seafood as part of a heart-healthy dietary pattern.
- Aim for approximately 2 servings of fish per week, especially fatty fish.

ALA

- Plant sources include walnuts, flaxseed, chia seeds, and canola or soybean oil.

Supplement counseling

- Routine over-the-counter fish-oil supplementation is not recommended for primary prevention of CVD.
- High-dose prescription EPA may be considered for selected high-risk patients with persistent hypertriglyceridemia under clinician supervision.
- Discuss bleeding risk, atrial fibrillation risk, product quality, and drug interactions.

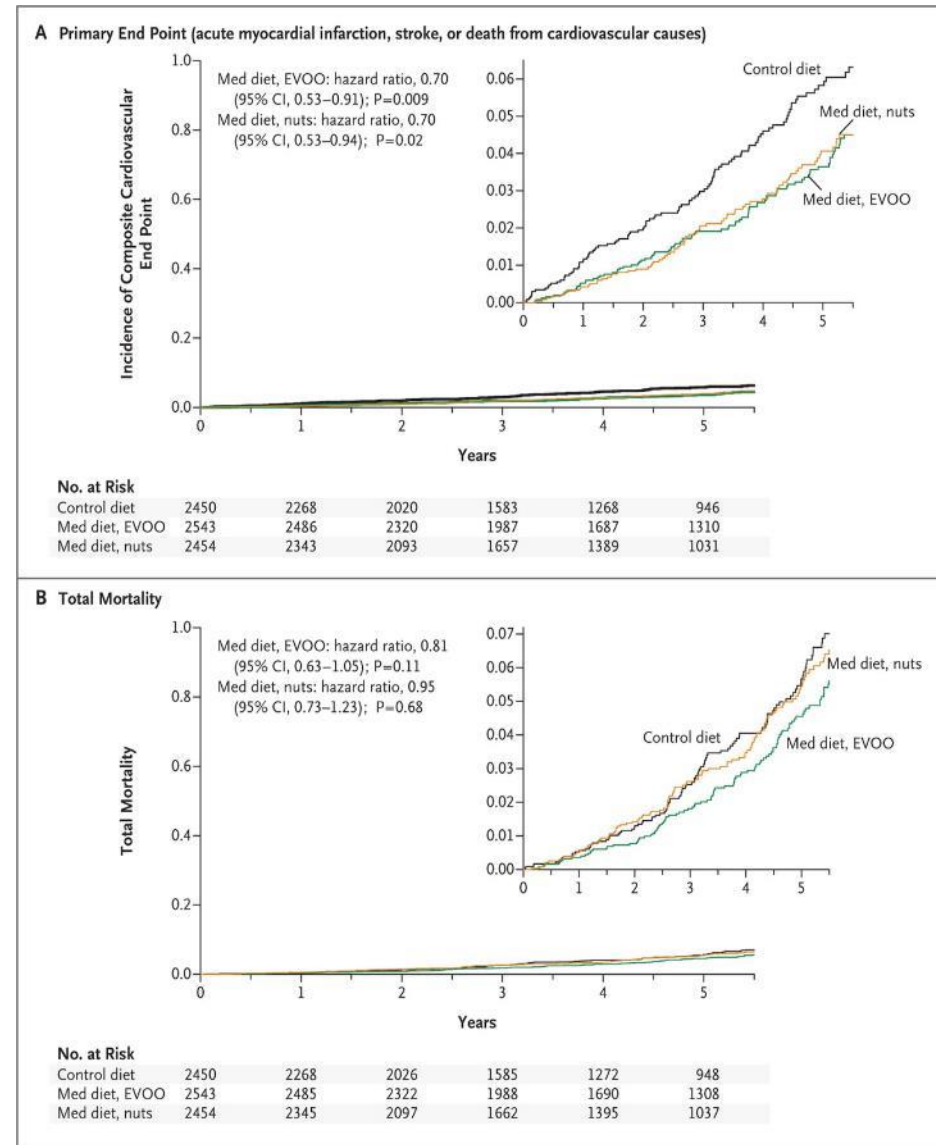
Monounsaturated Fatty Acids (MUFAs)

- **Basis of Mediterranean Diet:**
 - High in MUFA and low in SFA
→ decreased risk of CVD.
 - Emphasizes fruits, root vegetables, flax seeds, and **MUFAs**.
 - *Avocado, olives, nuts, olive oil and canola oil.*



Mediterranean Diet

- Primary Prevention of Cardiovascular Disease with a Mediterranean Diet (PREDIMED study)
- Among persons at **high cardiovascular risk**, a Mediterranean diet supplemented with extra-virgin olive oil or nuts **reduced the incidence** of major cardiovascular events.



Nutritional Guidance for Managing Hypertension

- **Consume:**

- Vegetables, fruits, whole grains
- Low-fat dairy products
- Poultry, fish, legumes, vegetable oils, nuts

- **Limit:**

- Sweets and sugar-sweetened beverages
- Red meats, saturated and trans-fats
- **Sodium (salt)**

Lower Sodium Intake

- Most dietary sodium comes from packaged, prepared, and restaurant foods, not the salt shaker.
- Clinical target: less than 2,300 mg sodium/day for most adults.
- Greater reductions may further lower blood pressure for many patients when nutritionally appropriate and feasible.

Practical strategies:

- • Compare Nutrition Facts labels
- • Choose “no salt added” or lower-sodium options
- • Flavor foods with herbs, spices, citrus, or vinegar
- • Use DASH-style eating patterns

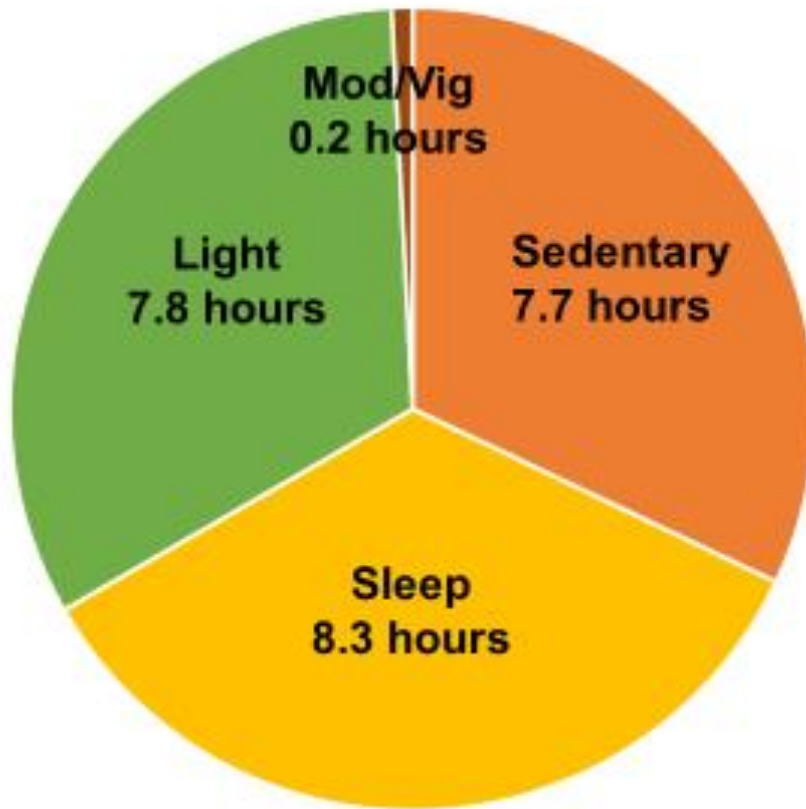
Combine DASH Diet With Lower Sodium Intake

- **DASH** = Dietary Approach to Stop Hypertension
 - **High** in vegetables, fruits, low-fat dairy products, whole grains, poultry, fish, and nuts.
 - **Rich** in potassium, magnesium, calcium, lean protein, and soluble fiber.
 - **Low** in sweets, sugar-sweetened beverages, and red meats.
 - **Low** in saturated fat, total fat, and cholesterol.

Steps to Promote Optimal Weight and Nutrition

- Goal: Reduce ASCVD risk
 - Measure BMI and waist circumference.
 - Discuss findings with the patient.
 - Assess patient's motivation to change.
 - Assess dietary intake.
 - 24-hour recall of their diet
 - Assess other lifestyle choices:
 - Tobacco use
 - Alcohol use
 - Physical activity level
 - Counsel patient about nutrition and set small goals to help them reach the main goal

Hours Per Day Spent in Various States of Activity



U.S. adults spend >7 h/d on average in sedentary activities. Replacing sedentary time with other physical activity involves increasing either moderate- to vigorous-intensity physical activity or light-intensity physical activity.

Key Takeaways

- Counsel to adopt a sustainable heart-healthy dietary pattern across the life course.
- Emphasize:
 - Vegetables and fruits, whole grains, legumes, nuts, healthy protein sources, and unsaturated fats
 - Minimally processed foods and culturally appropriate choices
- Minimize:
 - Ultra-processed foods, added sugars, excess sodium, trans fat, and foods high in saturated fat
- Prefer fish/seafood over routine fish-oil supplements for primary prevention.
- Use shared decision-making, small achievable goals, and follow-up to support behavior change.